

CLOQUET CARLTON COUNTY AP, MN, USA

WMO: 726558

Lat: 46.703N

Lon: 92.504W

Elev: 1279

StdP: 14.03

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04914

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.6	-12.2	-25.4	1.4	-16.9	-19.6	2.0	-10.0	23.7	13.4	20.8	11.0	7.2	290	0.609

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.7	84.1	70.4	81.6	67.7	79.2	65.9	74.0	80.8	71.3	78.4	68.9	75.7	9.5	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.1	124.7	78.9	69.6	114.1	75.8	66.1	101.0	72.9	38.4	81.1	35.9	79.4	33.9	76.6	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.9	18.7	16.7	DB	-24.0	90.2	6.5	2.4	-28.7	92.0	-32.4	93.4	-36.1	94.7	-40.8	96.5	
			WB	-24.1	77.4	6.3	2.5	-28.6	79.2	-32.3	80.7	-35.8	82.1	-40.4	83.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	40.2	11.0	14.9	26.7	39.0	50.6	59.9	66.4	64.5	56.6	44.1	29.9	16.9
	DBStd	21.41	13.41	12.75	12.06	8.69	7.74	6.84	5.98	5.76	7.95	8.35	10.72	11.98
	HDD50	5234	1208	984	724	342	86	6	0	0	27	225	605	1027
	HDD65	9273	1673	1404	1187	780	452	180	56	80	268	649	1052	1492
	CDD50	1649	0	0	2	13	104	303	508	449	227	41	2	0
	CDD65	211	0	0	0	0	4	27	99	63	18	0	0	0
	CDH74	2026	0	0	0	5	108	339	911	510	144	9	0	0
	CDH80	443	0	0	0	0	21	70	224	107	21	0	0	0
Wind	WSAvg	7.2	7.8	7.8	7.6	8.4	7.7	6.5	6.0	5.8	6.4	7.4	7.4	7.3
Precipitation	PrecAvg	30.9	1.1	0.8	1.6	2.3	3.3	4.2	4.1	4.0	3.9	2.7	1.9	1.1
	PrecMax	42.5	4.2	2.4	4.3	8.0	8.0	8.5	7.7	9.1	10.6	7.9	7.7	2.8
	PrecMin	20.0	0.1	0.2	0.2	0.2	0.4	0.4	0.9	0.5	0.6	0.3	0.1	0.2
	PrecStd	5.2	0.8	0.6	0.9	1.5	1.5	1.9	1.7	2.2	2.1	1.9	1.5	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.0	46.2	62.7	72.4	82.4	87.6	90.0	88.1	82.3	73.4	61.2	43.1
		MCWB	36.5	39.5	52.0	53.1	62.7	73.9	73.8	73.1	68.9	58.8	51.0	36.7
	2%	DB	36.2	39.5	53.8	64.5	76.9	81.5	84.4	82.2	78.9	66.4	54.2	38.5
		MCWB	33.4	35.6	44.9	49.3	58.8	67.2	71.3	69.4	67.0	56.7	48.0	35.5
	5%	DB	32.2	36.3	46.3	60.6	71.9	78.6	81.9	79.4	73.4	62.7	48.1	35.8
		MCWB	31.1	33.6	40.5	46.8	56.7	64.5	68.5	66.8	64.1	54.0	43.0	33.9
	10%	DB	28.4	33.2	42.6	54.5	66.0	73.6	79.4	75.5	70.4	56.9	45.0	33.5
		MCWB	27.3	31.0	37.1	43.9	54.4	62.4	66.9	65.6	62.6	50.2	40.8	32.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.3	41.0	53.4	57.4	66.6	75.5	77.4	76.7	73.2	63.9	54.3	38.8
		MCDB	41.5	44.7	61.4	64.9	74.6	82.9	85.0	83.5	80.0	67.6	57.2	40.7
	2%	WB	33.8	36.8	45.9	51.8	62.5	71.0	74.6	72.7	69.4	58.9	48.0	36.1
		MCDB	35.7	39.6	50.6	62.6	71.4	78.8	81.6	79.2	75.4	65.4	53.2	38.2
	5%	WB	30.9	33.9	41.2	47.8	59.2	67.5	71.9	69.9	66.6	54.8	44.3	33.5
		MCDB	32.3	36.2	46.5	58.1	68.6	74.4	78.9	76.6	71.8	60.4	47.5	35.0
	10%	WB	27.9	30.3	37.4	44.8	55.8	64.4	69.2	67.6	63.8	51.6	40.8	31.3
		MCDB	28.5	32.4	41.5	52.6	64.4	70.4	76.4	74.0	68.1	56.3	44.5	32.8
Mean Daily Temperature Range	5% DB	MDBR	16.3	17.6	17.7	19.2	21.8	21.2	20.7	19.5	19.0	16.0	13.8	13.5
		MCDBR	15.9	18.0	22.2	30.5	31.4	27.3	23.7	23.2	23.6	23.2	19.8	15.2
		MCWBR	13.5	13.9	14.6	17.4	16.4	14.9	12.2	12.2	14.0	14.1	13.5	12.2
	5% WB	MCDBR	15.1	15.1	21.1	27.3	26.7	23.3	22.7	20.4	20.2	20.3	17.9	13.4
		MCWBR	13.1	12.0	14.6	17.1	16.3	15.5	13.9	12.8	13.6	14.0	13.4	11.3
Clear-Sky Solar Irradiance	taub	0.242	0.255	0.301	0.346	0.375	0.369	0.377	0.382	0.356	0.308	0.283	0.258	
	taud	2.400	2.374	2.377	2.375	2.343	2.384	2.385	2.391	2.438	2.566	2.524	2.423	
	Ebn at Noon	278	296	294	289	282	284	280	274	272	273	260	257	
	Edh at Noon	22	27	32	36	38	37	36	35	30	23	19	19	
All-Sky Solar Radiation	RadAvg	491	847	1203	1475	1687	1829	1951	1634	1231	751	460	347	
	RadStd	54	74	115	144	127	126	80	125	92	100	49	33	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	-1.62	-2.07	N/A	N/A	N/A	N/A	
(47)	Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page